

Bitcoin On-Chain Data Warehouse

From Raw Blockchain to Multidimensional OLAP Analysis

657K+

Transactions

4

Dimensions

10

OLAP Queries

7/10

Queries Solved

Presentation Structure

01 Project Goal & Dataset →

02 Solution Architecture →

03 ETL Process Design →

04 ETL Implementation →

05 Dimensional Model →

06 OLAP Analysis (10 Queries) →

07 Management Dashboard →

08 Summary →

What We're Building - and Why

Core Research Question

"Has Bitcoin successfully evolved from a low-fee experimental payment system into a **high-value settlement layer** driven by market sentiment and protocol upgrades?"

ON-CHAIN

mempool.space API

Blocks: height, miner pool, weight, fees
Txs: inputs/outputs, SegWit, RBF, locktime

MARKET PRICE

Yahoo Finance API

Daily BTC-USD OHLCV
(Open / High / Low / Close / Volume)

SENTIMENT

Alternative.me API

Fear & Greed Index (0–100)
Available from 2018-02-01

DOMINANCE

CoinLore API

BTC market cap dominance %
vs total crypto market

TARGET USER

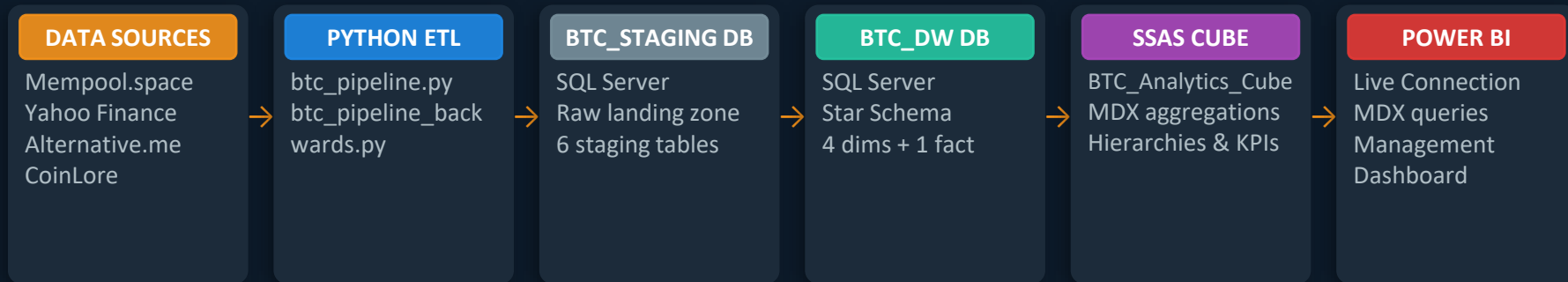
Blockchain Analyst / Researcher

Monitors on-chain settlement dynamics, fee pressure, and market-sentiment correlations to evaluate Bitcoin network capacity, protocol adoption, and macro-level valuation.

KEY QUESTIONS THIS DW ANSWERS

- Do people overpay fees when the market is gripped by fear?
- Is Tuesday genuinely the most expensive day to send Bitcoin?
- Has SegWit adoption changed how efficiently the network processes value?

Full-Stack Architecture



TOOL JUSTIFICATIONS



Python over SSIS/Talend

Mempool API returns HTTP 404 (not empty list) when a block has exactly 25/50 txs. Python's custom `http_get(ignore_404=True)` solves this; SSIS cannot handle it gracefully. Also, because I love Python.



Staging + DW Separation

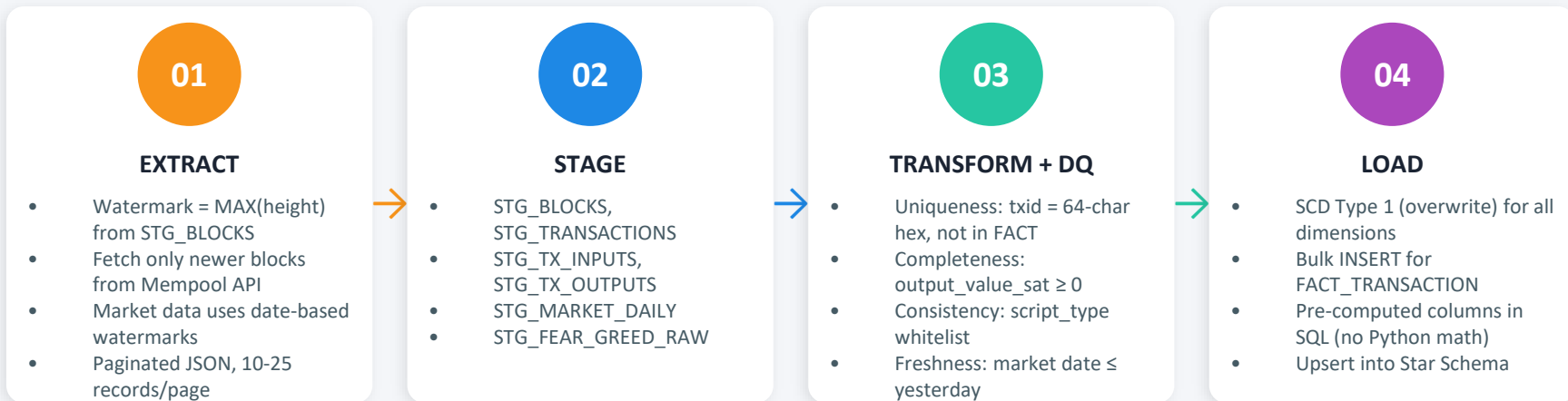
Staging preserves raw data intact for re-runs. DW contains only clean, conformed data. An ETL bug never corrupts the analytical layer.



SSAS Live Connection in Power BI

657,968 rows (growing via backfill).
Live = Power BI sends MDX queries to pre-aggregated cube, never loading raw rows into RAM → sub-second dashboard response.

ETL Pipeline: 4-Phase Design



DATA QUALITY GATES

Gate	Rule	On Failure
Uniqueness	txid = 64-char hex, not already in FACT	Log to ERR_QUALITY_REJECTS, skip row
Completeness	output_value_sat ≥ 0, block_hash resolves to DIM_BLOCK	Reject entire block batch
Consistency	script_type ∈ {P2PKH, P2SH, P2WPKH, P2WSH, P2TR, OTHER}	Map to 'OTHER', log
Freshness	Staging market date must be ≤ yesterday	Alert logged, load continues

Technical Highlights & Bug Fixes

AUTONOMOUS BACKWARD BACKFILL

btc_pipeline_backwards.py

1. Queries MIN(height) from staging → finds history gap
2. Fetches blocks in batches of 200, walking toward block 0
3. Commits each batch fully before starting the next
4. Resumable: Ctrl+C stops safely - restart picks up exactly where it left off
5. Self-healing: 3 consecutive empty batches trigger a safety exit (anti-infinite-loop guard)

HARD BUGS SOLVED



Arithmetic Overflow

Problem: pyodbc inferred satoshi values as tinyint (max 255 vs 2.1×10^{15} sat)

→ Fix: Added **CAST(? AS BIGINT)** in every INSERT



Msg 1934 DDL Error

Problem: QUOTED_IDENTIFIER OFF prevents computed column creation

→ Fix: **SET QUOTED_IDENTIFIER ON** in all DDL scripts



404 Pagination Crash

Problem: Mempool API returns 404 (not empty list) at page boundary (25/50 txs)

→ Fix: **ignore_404=True** param in **http_get()** returns [] and breaks loop



NVT/Fear-Greed Not Loading

Problem: load_ts not bumped when staging updated → DW thought data unchanged

→ Fix: **Always update load_ts after merge, forcing SCD-1 UPDATE to propagate**

Star Schema: FACT_TRANSACTION

DIM_DATE

SCD Type 1

- Year / Quarter / Month / Day
- Day of Week
- Halving Era (1–5)
- Difficulty Tier

DIM_BLOCK

SCD Type 0 (Immutable)

- Block Height
- Hash
- Miner Pool Name & Slug
- Total Size, Weight, Tx Count
- Difficulty

FACT_TRANSACTION

Grain: one row per transaction

- fee Currencies (Sat, Btc, Usd)
- Transaction Count
- input/output value (Sat, Btc, Usd)
- fee_usd
- fee_burden_pct
- Tx Size Bytes
- Tx Vsize Bytes
- Tx Weight Units

DIM_MARKET

SCD Type 1

- Price / Market Cap
- Fear & Greed Score (0–100)
- Fear & Greed Label (text)
- BTC Dominance %, NVT Ratio

DIM_TX_TYPE

Junk Dimension - SCD Type 1

- script_type_desc
- segwit_flag
- coinbase_flag
- rbf_flag, locktime_flag

10 Queries - Status Overview

7

Fully Answered

2

Partially Answered

3

Structurally Impossible

Era 5

Data Coverage

✓ SOLVED

Q1.1 - SegWit Adoption Rate by Quarter

89.62% SegWit-native in 2026 Q2

✓ SOLVED

Q1.2 - Fee-Rate Spikes Under Network Congestion

June: higher fee rate (18.97) but lower fee burden (6.14%)

✗ N/A

Q1.3 - Fee Burden by Transaction Size Tier

No DIM_SIZE_TIER dimension → impossible without DAX bucket

✓ SOLVED

Q1.4 - Tx Volume vs. Fear & Greed Index

Extreme Fear dominates: 370K txs, \$205.5B output value

✗ N/A

Q1.5 - Multi-Input vs. Single-Input Structure

input_count exists in staging but not promoted to FACT

~ PARTIAL

Q1.6 - Transaction Evolution Across Halving Eras

Only Era 5 populated; backfill in progress

✓ SOLVED

Q1.7 - Coinbase vs. Standard Composition

Coinbase = 1% by count, 0.40% by BTC value - subsidy minimal in Era 5

✗ N/A

Q1.8 - NVT Ratio Under Sentiment Regimes

Market cap approximated + only 47 days of data → insufficient

~ PARTIAL

Q1.9 - Block Space Efficiency by Difficulty Tier

All txs in single 'Extreme' tier - cross-tier comparison impossible now

✓ SOLVED

Q1.10 - Weekend vs. Weekday Fee Patterns

Sunday = cheapest (14.57 sat/vB) · Tuesday = peak (24.07 sat/vB)

06 · OLAP ANALYSIS

Q1-SegWit Adoption by Quarter

Year	SegWit Share %	Transaction Count
2026	89,62	651368
2	89,62	651368
OTHER	110,88	526462
P2PKH	859,95	67883
P2SH	1.023,72	57023
Total	89,62	651368

Q2-Fee Rate Spikes

Year	Avg Fee Rate	Avg Fee Burden %	Avg IO Value Ratio
2026	18,26	8,55	1,0753
5	17,68	10,55	1,0952
6	18,97	6,14	1,0512
Total	18,26	8,55	1,0753

Q4- Volume vs Fear & Greed

Fear Greed Label	Transaction Count	Output Value Usd
Extreme Fear	370501	205.516.584.673,10
Fear	235802	132.589.062.131,47
Neutral	51665	27.966.838.070,49
Total	657968	366.072.484.875,06

Q6-Halving Eras

Halving Era	Transaction Count	Tx Vsize Bytes
Era 5	657968	170.324.696,00
Total	657968	170.324.696,00

Q7-Coinbase vs Standard

Halving Era	Transaction Count	Coinbase Share %	Output Value Btc
Era 5	657968		5.140.367,38
No	651368		5.119.597,72
Yes	6600		20.769,66
Total	657968		5.140.367,38

Q9-Block Space Efficiency

Difficulty Tier	Tx Vsize Bytes	Avg Fee Rate	Transaction Count
Extreme	170.324.696,00	18,26	657968
Total	170.324.696,00	18,26	657968

Q10-Weekend vs Weekday

Day Name	Avg Fee Rate	Transaction Count	Output Value Usd
Friday	18,71	101815	61.599.998.700,42
Monday	17,38	102221	60.932.072.940,97
Saturday	15,17	102654	43.363.802.526,28
Sunday	14,57	95680	55.463.275.558,31
Thursday	18,82	95015	54.074.502.935,78
Tuesday	24,07	81001	41.961.685.627,87
Wednesday	20,68	79582	48.677.146.585,44
Total	18,26	657968	366.072.484.875,06

5 Standout Results from Live SSAS Cube

Q1.4

Fear = More Activity

Counter-intuitive: Extreme Fear days account for 370,501 txs (\$205.5B output). No Greed/Extreme Greed observed - market was in persistent fear throughout Era 5 Q2 2026. Consistent with panic-selling or accumulation during downturns.

Q1.10

Tuesday = Most Expensive Day

Avg fee rate: Tuesday 24.07 sat/vByte · Sunday 14.57 · Saturday 15.17. Institutional settlement concentrated on business days creates a 30–40% weekend fee discount. Volume peaks Saturday (102,654 txs) and Monday (102,221).

Q1.1

SegWit Already Won

89.62% of 651,368 non-coinbase transactions in Q2 2026 are SegWit-native. P2PKH (legacy) is now the minority segment with 67,883 txs. Full S-curve trajectory not yet visible (only Era 5 loaded).

Q1.2

Fewer Txns = Efficient Fee Absorption

May 2026: 10.55% avg fee burden. June 2026: fee rate rose to 18.97 sat/vByte but fee burden fell to 6.14%. Fewer but higher-value txs in June absorbed fees more efficiently. IO Value Ratio $\approx$ 1.05–1.10 both months.

Q1.7

Coinbase Share is Minimal in Era 5

6,600 coinbase txs out of 657,968 total (1.00% by count). Output value: coinbase = 20,769 BTC vs standard = 5.12M BTC (0.40% share). Confirms block subsidy is now economically marginal - fee revenue is the emerging security model.

Power BI Dashboard - Live SSAS Connection

01

Strategic Anchor

Network Fee Sustainability KPI Card
(traffic-light: Red/Yellow/Green)

02

Historical Evidence

Settlement vol + Fee rate over time
SegWit adoption trajectory (100% stacked bar)

03

Structural Drivers

Sentiment-volume scatter · Difficulty tier bar
Miner revenue donut · Day-of-week heatmap

7 DASHBOARD VISUALS

1

Card

Network Fee Sustainability Score

→ Red (0.00) - subsidy still dominant in Era 5

2

Dual-Axis Line

Settlement Volume vs. Avg Fee Rate over Time

→ May→June: output fell, fee rate rose
(divergence pattern)

3

100% Stacked Bar

SegWit vs. Legacy Script Adoption by Quarter

→ 89.6% SegWit-native in Q2 2026

4

Scatter Plot

F&G Score vs. Daily Settlement Volume

→ Extreme Fear days cluster highest output values

5

Horizontal Bar

Avg Fee Burden (%) by Difficulty Tier

→ 8.55% for sole 'Extreme' tier (single-tier dataset)

6

Donut Chart + Filter Era

Output Value BTC by Script Type

→ P2PKH = 84.1% of BTC value (4.32M BTC)

7

Matrix

Avg Fee Rate: Day of Week × Halving Era

→ Sunday cheapest · Tuesday most expensive · actionable!

0,00

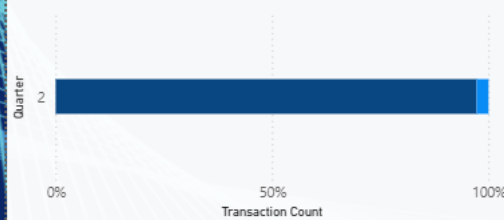
Network Fee Sustainability

Settlement Volume vs. Avg Fee Rate

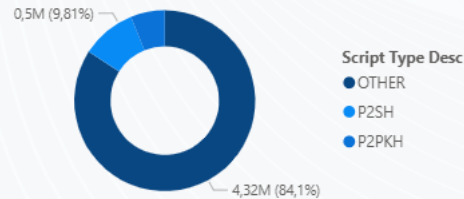


Transaction Count by Quarter and Script Type Desc

Script Type Desc ● OTHER ● P2PKH ● P2SH



Output Value Btc by Script Type Desc



Fear & Greed Score vs. Daily Settlement Volume



Avg Fee Burden (%) by Difficulty Tier



Halving Era

- ☐ Era 1
- ☐ Era 2
- ☐ Era 3
- ☐ Era 4
- ☐ Era 5

Avg Fee Rate by Day of Week x Halving Era

Day Name	Era 5	Total
Friday	18,71	18,71
Monday	17,38	17,38
Saturday	15,17	15,17
Sunday	14,57	14,57
Thursday	18,82	18,82
Tuesday	24,07	24,07
Wednesday	20,68	20,68
Total	18,26	18,26

Project Outcomes & Next Steps

✓ WHAT WAS ACHIEVED

- Full ETL pipeline: 2 Python scripts, 6 staging tables, 4 DW dimensions, 1 fact table
- SSAS multidimensional cube + Power BI live connection dashboard
- 657,968 transactions loaded (and growing via backfill)
- 10 OLAP queries: 7 answered with live results, 3 correctly identified as structurally impossible
- Unanswerable queries documented with precise technical justifications - not silently omitted

! PRIMARY LIMITATION

- Dataset covers only Era 5 (May–June 2026) - cross-era comparisons analytically limited with a single era
- Several dashboard visuals trivial: difficulty tier bar (1 tier), heatmap (1 era column)
- Fix underway: `btc_pipeline_backwards.py` downloading historical blocks toward Era 1 - all cross-era queries auto-populate as it progresses

V2 MODEL ADDITIONS

1. Add `input_count` to `FACT_TRANSACTION` → unlocks Q1.5
2. Add `DIM_SIZE_TIER` dimension → unlocks Q1.3
3. Add circulating supply fact table → enables precise NVT

2

Python Scripts

6

Staging Tables

5

DW Tables

1

SSAS Cube

7

Power BI Visuals

Thank You

Questions & Discussion

657K+

Transactions
Loaded

7 / 10

OLAP Queries
Answered

Era 5

Data Coverage
(backfill ongoing)

Live

SSAS Cube
Power BI Connection